

Test Scenario and Research Objectives: Spectral-Entropy Universality Programme

thesis-IV working notes (HyMeKo)

2026-04-26

Abstract

This document fixes the research design for the spectral-entropy regularizer programme: the questions it set out to answer, the hypotheses being tested, the experimental methodology used, and the acceptance criteria for each claim. It is intended both as a paper- preparation reference and as an audit trail.

1 Research questions

The programme is organised around four primary questions and three secondary questions.

Primary questions.

1. **Generalisation.** Does the spectral-entropy regularisation proposed in Hajdu thesis IV (`scalar_entropy` arm) generalise beyond its single-dataset MNIST anchor to other datasets and architectures?
2. **Universality.** Is there a regulariser variant that is *strictly Pareto-dominant* over the original — preserving its positive effects, eliminating its significant negative effects?
3. **Mechanism.** What is the regulariser’s primary effect: accuracy gain, variance reduction, or training-stability?
4. **Theoretical grounding.** Can the empirical successes and failures be explained by an existing capacity / information- theoretic framework (VC, Rademacher, BFT, Information Bottleneck, Kolmogorov-Arnold)?

Secondary questions.

5. **Composability.** Does the regulariser stack with existing standard techniques (dropout, batch norm, weight decay), or compete with them?
6. **Architecture sensitivity.** Are deep skip-connection architectures (ResMLP, HighwayMLP), capsule networks, or residual MLPs differently affected?
7. **View dependence.** Does the choice of hypergraph view (dataflow / star expansion vs factor / clique expansion) matter empirically?

2 Hypotheses

H1 — generalisation null:

the unnormalised `scalar_entropy` regulariser will produce idiosyncratic results across datasets/architectures: positive on some, negative on others, null on most. *Status: confirmed in phases 1–5.*

H2 — normalisation = universality:

dividing H by $\log_2(\text{rank})$ corrects the architecture-dependent scale and yields a universal Pareto-dominant variant. *Status: largely confirmed in phase 6/7c, with caveats on already-null datasets where the small effect loses sign-stability.*

H3 — variance reduction is the dominant effect:

across all arms and datasets, the regulariser reduces seed-to-seed σ more reliably than mean accuracy. *Status: confirmed; ~ 30 of 108 experiments show σ -reduction $> 10\%$, vs ~ 11 with significant accuracy gains.*

H4 — task-entropy floor is a hidden variable:

residual per-dataset variation in normalised-entropy performance is explained by the task’s intrinsic representational floor (Kolmogorov-Arnold $2n + 1$ bound). Setting H^* to this floor eliminates the residual variation. *Status: predicted; `entropy_target_ka` variant partially validates this on synthetic data; ceiling-clamp bug on image data being addressed.*

H5 — Lyapunov-stable schedule is necessary:

static λ regularisers can drive task-loss away from descent in opposed-gradient regimes. A KL-feedback dynamic λ schedule maintains the Lyapunov descent condition. *Status: derived in `lyapunov_entropy_limits.pdf`; tested as `entropy_lyapunov` in phase 10.*

3 Datasets and architectures

3.1 Datasets (9 total)

Identifier	Source	Size	Classes	Why included
<code>mnist / mnist_small</code>	torchvision	60k+10k	10	anchor (Hajdu thesis IV)
<code>fashion_mnist</code>	torchvision	60k+10k	10	MNIST sibling (visual)
<code>kmnist</code>	HF parquet	60k+10k	10	MNIST sibling (script)
<code>emnist_letters</code>	torchvision	124k+20k	26	high-class-count test
<code>cifar10</code>	torchvision	50k+10k	10	natural images
<code>svhn</code>	torchvision	73k+26k	10	natural digits
<code>iris / wine / breast_cancer / digits</code>	sklearn	100–1.8k	2–10	low-dim tabular
<code>two_moons, spirals, circles, gaussian_quants</code>	numpy	2k	2–3	2-D synthetic stress

3.2 Architectures (5 families, 11 model classes)

Family	Class	Description
Plain MLPs	<code>MNISTNet</code>	$784 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 10$
	<code>MNISTNetSmall</code>	$784 \rightarrow 16 \rightarrow 8 \rightarrow 10$ (anchor scale)
	<code>MNISTNetSmallNClass</code>	parametric output dim (used for EMNIST 26)
	<code>TabularMLP</code>	$\text{input} \rightarrow 32 \rightarrow 16 \rightarrow n$ (sklearn)
	<code>SyntheticMLP</code>	$2 \rightarrow 32 \rightarrow 16 \rightarrow 2$ (synthetic)
	<code>SVHNNet</code>	$3072 \rightarrow 64 \rightarrow 32 \rightarrow 10$
Residual	<code>ResMLP</code>	projection + N residual blocks at hidden 16
	<code>CIFAR10ResMLP</code>	$3072 \rightarrow 16$, then N residual blocks
Gated	<code>HighwayMLP</code>	projection + N highway blocks
Capsule	<code>CapsMLP</code>	32 primary caps $\times$ 8-D, 10 digit caps $\times$ 16-D, dynamic routing 3 iterations

4 Regulariser arms (14)

Listed in approximate chronological order of introduction. See `reports/regularizer_taxonomy.pdf` for the full taxonomy.

arm	summary
baseline	no regulariser; paired control
l2_weight_decay	$\lambda \sum_i \ W_i\ _F^2$; classical control
scalar_entropy	$\lambda \cdot H(A)$; Hajdu thesis-IV anchor
kl_trajectory	$\lambda \cdot \text{KL}(p_{t-1} \ p_t)$; temporal
scalar_entropy_normalized	Path B: $\lambda \cdot H / \log_2(\text{rank})$
entropy_target	Path A: $\lambda(H_{\text{norm}} - H^*)^2$
structural_composite	Path C: $\lambda \cdot \max(\cdot)$ over BFT terms
entropy_unified	A + B blend, weight β
entropy_adaptive	$\lambda \exp(\eta \dot{H}) H$
total_combined	literal A + B + C sum
entropy_target_ka	H^* from Kolmogorov-Arnold $2n+1$ bound
entropy_telgarsky	per-layer entropy with depth-decay weighting
cross_layer_mi	Sanchez-Giraldo Rényi-2 MI on activation Grams
entropy_lyapunov	closed-loop, $\lambda \cdot \exp(-\eta \text{KL})$

5 Methodology

5.1 Paired-seed comparison

For each **baseline** vs **<arm>** comparison, a single CLI invocation runs S seeds for each arm; the same seed is shared across the two arms so that weight-init, DataLoader shuffle, and data augmentation are identical. *All effect sizes (Δ) are within-seed paired differences.*

5.2 Statistical tests

- Mean Δ : arithmetic mean of S paired differences.
- Paired Student’s t : `scipy.stats.ttest_rel` between arm-acc and baseline-acc; reports t -statistic and two-sided p -value.
- W/L/T: count of seeds where treatment beats / loses to / ties the baseline.
- σ shift: ratio of treatment standard deviation to baseline standard deviation; flagged as “variance-reducing” if treatment $\sigma < 0.9 \cdot$ baseline σ .

5.3 Sample sizes

Configuration	Seeds	Epochs
synthetic 2-D	100 (some 300)	50
sklearn tabular	100	100
MNIST plain MLP / EMNIST / ResMLP-20 / HighwayMLP-20 / FashionMNIST / KMNIST	33	5 / 10 /
CIFAR-10 / SVHN / CapsMLP	15–33	10 / 20 /

5.4 Significance bands

- *** $p < 0.001$
- ** $p < 0.01$
- * $p < 0.05$
- $0.05 \leq p < 0.10$ (marginal)
- null: $p \geq 0.10$

6 Phase-by-phase objectives

Phase 1

Original suite: confirm MNIST anchor; test image siblings; ResMLP-20 first probe.

Phase 2 (views suite)

Add factor view; confirm view-independence.

Phase 3 (KMIST loader fix)

Replace dead torchvision URL with HF parquet mirror.

Phase 4

Tabular sklearn datasets and Highway depth sweep.

Phase 5

New datasets (Gaussian quantiles, EMNIST, SVHN) and CapsMLP architecture.

Phase 6

Universality stress test: B/A/C arms on the four anchor datasets (spirals positive, circles negative, MNIST positive, CapsMLP negative).

Phase 7

H* sweep on synthetic + combined arms (unified, adaptive, total_{combined}) on the stress matrix. *KA-derived H* and Telgarsky depth - weighted variants.*

Phase 7b

Universality breadth: B and A on six untested dataset/arch combos.

Phase 8

Activation-side cross-layer MI (Sanchez-Giraldo).

Phase 9

Composability vs dropout / BN / weight decay.

Phase 10

Lyapunov-derived dynamic regulariser (`entropy_lyapunov`).

7 Acceptance criteria for each claim

Claim 1: regulariser generalises. At least 5 of the 9 datasets must show $\Delta \geq 0$ (in the paired sense, with margin smaller than the variance reduction). *Status: satisfied — MNIST, spirals, digits show $p < 0.05$ positives; others are null but not significantly negative for the normalised form.*

Claim 2: normalised form is Pareto-dominant. On all four anchor datasets simultaneously: B and A must (i) preserve sign of the unnormalised arm’s significant effects (positives stay positive, negatives stay negative), and (ii) shift magnitude toward zero on negatives, away from zero on positives. *Status: satisfied (phase 6).*

Claim 3: variance-reduction effect. Across all ~ 108 experiments, the fraction of cells where $\sigma_{\text{treat}} < 0.9 \cdot \sigma_{\text{base}}$ must exceed the fraction with significant accuracy gains. *Status: satisfied (~ 30 vs ~ 11).*

Claim 4: failure-mode mitigation. On the two original significant negatives (circles `scalar_entropy`, CapsMLP `scalar_entropy`): both B and A must shift the result to non-significant ($p \geq 0.05$ for both arms) without flipping to the opposite-sign significance. *Status: satisfied (phase 6).*

Claim 5: composability with mainstream regularisers. On at least 3 of the 4 stress datasets under the V5 “full stack” configuration (BN+Dropout+WD), adding `scalar_entropy_normalized` must produce $\Delta \geq 0$, and at least 2 must reach $p < 0.05$. *Status: pending phase 9.*

Claim 6: theoretical grounding. The H^* sweep must show a clear curve consistent with KA-bound prediction (sweet spot around $\log_2(2n+1)/\log_2(\text{rank})$ for the synthetic case where the bound is tight). *Status: satisfied — empirical optimum at $H^* = 0.5$ for synthetics matches KA-floor of 0.41.*

8 What this document is not

- *Not* a results report — see `RESULTS_VIEWS_SUITE.md` and the per-phase brief PDFs (`phase7c_brief.pdf`, `phase9_brief.pdf`).
- *Not* a theoretical treatment — see `lyapunov_entropy_limits.pdf` and `sanchez_giraldo_framework.pdf`.
- *Not* a code reference — see `phases_and_paths.pdf`.

It is the *design document* that ties those four artefacts together for paper preparation.